

FOORS⁺ over Bluetooth Low Energy: A Low-Latency Opportunistic Routing and Dissemination Algorithm

Luca Ilchen, Patrick Reich, Axel Herrmann

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Abstract

FOORS⁺ is a lightweight routing and dissemination algorithm for transmitting triage records over opportunistic Bluetooth Low Energy (BLE) encounters in disaster-response scenarios. The primary design objective is minimal latency to reach any sink under realistic constraints: short-lived links, costly connection establishment, and low stable node degree. This manuscript provides a mathematically precise algorithm definition aligned with the reference implementation in the `ble_foors_lib` codebase.

1 Motivation and Design Objective

In disaster-response settings, networking conditions are dominated by intermittent connectivity, short encounters, and highly variable paths. FOORS⁺ is designed under the following priorities:

Factor	Priority
Minimal latency to a sink	highest
Robustness to instability (changing paths, short encounters)	required
Energy efficiency	important but secondary
No central routing infrastructure	required
Minimal overhead and simple logic	desirable

These priorities yield three guiding principles: (i) connections are expensive (avoid active polling), (ii) sending data is an opportunity (use connections immediately), (iii) controlled flooding is practical at small degrees.

2 Notation and Conventions

We summarize the principal symbols used throughout the manuscript.

Symbol	Meaning
V	set of devices (nodes)
$E(t)$	connectivity edges at time t
$\Gamma(u, t)$	neighbors of node u at time t
S	set of sink nodes
$\text{id}(u)$	persistent node identifier of u (UUID-like)
τ	triage identifier (UUID)
P	triage payload (application map/object)
s	sink identifier (a value in Σ^*)
$s? \in \Sigma^* \cup \{\perp\}$	optional sink destination; $\perp$ denotes “non-addressed”
$R_u(s)$	neighbor hop-count table at u for sink s
$H_u(s)$	equal-cost next hop set toward sink s
$\mathcal{S}_u$	set of known sinks at u (keys of routing state)
$\text{Flood}(u)$	global flooding predicate at node u
T_u	local triage store at node u (map from τ)
$A_u(\tau)$	assumed-delivered flag for triage τ at u
$F_u(\tau)$	flooding-seen flag for triage τ at u

3 System Model

3.1 Network and time

Time is continuous $t \in \mathbb{R}_{\geq 0}$. The dynamic connectivity graph is

$$G(t) = (V, E(t)),$$

where V is the set of devices and $(u, v) \in E(t)$ iff u and v can exchange messages at time t . For $u \in V$, let the neighbor set be

$$\Gamma(u, t) = \{v \in V : (u, v) \in E(t)\}.$$

Figure 1 illustrates a representative encounter snapshot $G(t)$.

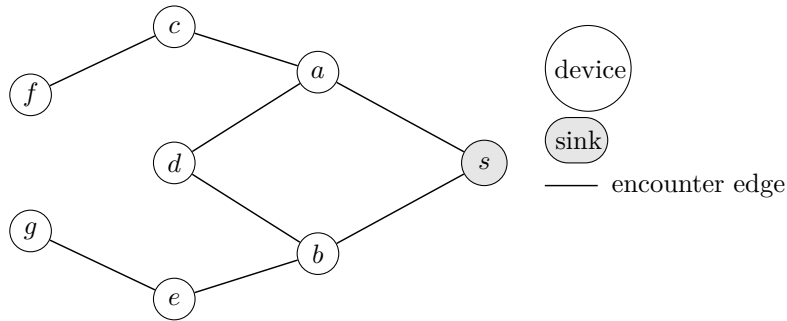


Figure 1: Example encounter snapshot $G(t) = (V, E(t))$ with one sink node s highlighted.

3.2 Identifiers

Each node $u \in V$ has a persistent identifier $\text{id}(u) \in \Sigma^*$ (UUID-like). A subset of nodes are sinks; let $S \subseteq V$ denote the set of sinks.

A triage record is identified by a globally unique identifier $\tau \in \Sigma^*$ (UUID; collisions are considered negligible) and carries an application payload P .

4 Message Types and Transport

4.1 Message types

FOORS⁺ uses two message types:

1. **RoutingUpdate** messages announce hop-count information per sink.
2. **Triage** messages carry triage data and an optional sink destination.

4.2 Encoding and chunking

Messages are encoded as CBOR and transported over BLE by fragmenting a logical message into at most 256 chunks. Each chunk contains a header (message UUID bytes, chunk index, total-chunks) followed by payload bytes. Let the header length be H bytes (implementation: $H = 18$). For MTU M , the maximum chunk payload is $M - H$.

4.3 Triage message body

A triage body is CBOR list

$$[\tau, P, s?],$$

where $s? \in \Sigma^* \cup \{\perp\}$. If $s? \in \Sigma^*$, the triage is *sink-addressed* to sink identifier s . If $s? = \perp$, the message is *non-addressed*. In the reference implementation, non-addressed triages are used for (i) triages that should reach all sinks, and (ii) opportunistic re-seeding on new connections.

4.4 RoutingUpdate message body

A routing update body is CBOR list

$$[s, h, x],$$

where s is the sink identifier, $h \in \mathbb{Z}$ is the hop count, and x is the sender identifier. The distinguished value $h = -1$ means route withdrawal.

5 Local State

5.1 Routing state

Each node u maintains a per-sink neighbor table:

$$R_u(s) : \Gamma(u) \rightarrow \mathbb{Z},$$

where $R_u(s)[v]$ is the hop count advertised by neighbor v for sink s . Missing entries mean “no route via that neighbor.”

Define the set of equal-cost next hops:

$$H_u(s) = \arg \min_{v \in \text{dom}(R_u(s))} R_u(s)[v].$$

If $R_u(s)$ is empty then $H_u(s) = \emptyset$.

Flooding-mode predicate (global). Let $\mathcal{S}_u$ be the set of known sinks (routing-table keys). FOORS⁺ defines a global flooding-mode predicate:

$$\text{Flood}(u) \iff (\mathcal{S}_u = \emptyset) \vee (\exists s \in \mathcal{S}_u : H_u(s) = \emptyset).$$

When $\text{Flood}(u)$ holds, routing mode is suppressed.

Sink forgetting. For each sink s , if $R_u(s)$ becomes empty a timer of duration T_{rm} is started. If no route reappears by timeout, s is removed from $\mathcal{S}_u$.

5.2 Triage state

Each node stores triages by ID. For each τ it stores: (i) payload P_τ , (ii) an *assumed-delivered* flag $A_u(\tau) \in \{\text{true}, \text{false}\}$, (iii) a *flooding-seen* flag $F_u(\tau) \in \{\text{true}, \text{false}\}$.

$A_u(\tau)$ is set to true if forwarding in routing mode reports success (at least one unicast send succeeded in the relevant routing step). This is not an end-to-end acknowledgement.

In the reference implementation, $F_u(\tau)$ is also used as a guard to avoid repeated flooding; notably, some failures (e.g., exceeding a retry bound without any successful routing transmission) may still set $F_u(\tau) = \text{true}$ even if no broadcast occurred.

6 Routing Layer

6.1 Connection established

When a connection to peer v is established, a sink node u sends a self-announcement for $\text{id}(u)$ with hop count 0. Additionally, for each known sink $s \in \mathcal{S}_u$, node u sends hop information derived from its best next hop, using split-horizon-like loop avoidance: if v is the current best next hop for s , u does not advertise s to v .

Figure 2 illustrates how hop-count knowledge toward a sink can evolve; unknown hop-counts are rendered as ∞ .

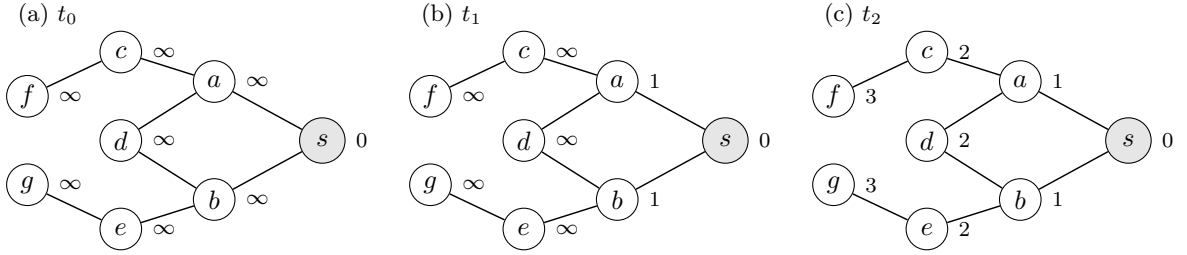


Figure 2: Illustration of hop-count knowledge toward sink s evolving via RoutingUpdate dissemination. Unknown hop-counts are denoted by ∞ .

6.2 RoutingUpdate receive

Upon receiving an update $\langle s, h, x \rangle$ from peer v : (i) ignore updates about self if in sink mode and $s = \text{id}(u)$, (ii) if $h = -1$, remove v from $R_u(s)$ and possibly enter flooding mode for s , (iii) if $h \geq 0$, set $R_u(s)[v] \leftarrow h$, publish sink discovery if s is new, and broadcast updated information if the best hop changes.

6.3 Connection lost

When peer v disconnects, remove v from all tables. If a sink becomes unreachable, broadcast withdrawal. If the best hop changes, broadcast updated hop count and exclude the current best-next-hop from receiving that broadcast. If the identity of the best next-hop changes, send a withdrawal update to the new next-hop.

7 Triage Dissemination Layer

7.1 First-sight delivery

When a node observes triage ID τ for the first time, it delivers the payload to the application callback.

7.2 Sink-local termination

If a triage is sink-addressed to $s = \text{id}(u)$, node u stores it and performs no forwarding.

7.3 Forwarding rules

Sink-addressed triage. If a triage is addressed to sink $s \neq \text{id}(u)$, forward in routing mode if $H_u(s) \neq \emptyset$ by sending to all equal-cost next hops. If $H_u(s) = \emptyset$, broadcast (flood) once per triage ID guarded by $F_u(\tau)$.

Non-addressed triage. If $s? = \perp$, treat the triage as destined to *all sinks*. If routing is possible (not flooding mode), the node forwards one copy toward each known sink; otherwise it broadcasts a non-addressed triage.

Opportunistic re-seeding. Upon connection establishment, a node sends all triages with $A_u(\tau) = \text{false}$ to the newly connected peer as non-addressed triages.

Sink-to-sink replication. When a sink learns of a new sink s (via sink discovery), it forwards all stored triages toward s .

7.4 Routing Flowchart

Figure 3 summarizes the triage handling logic in a compact flowchart.

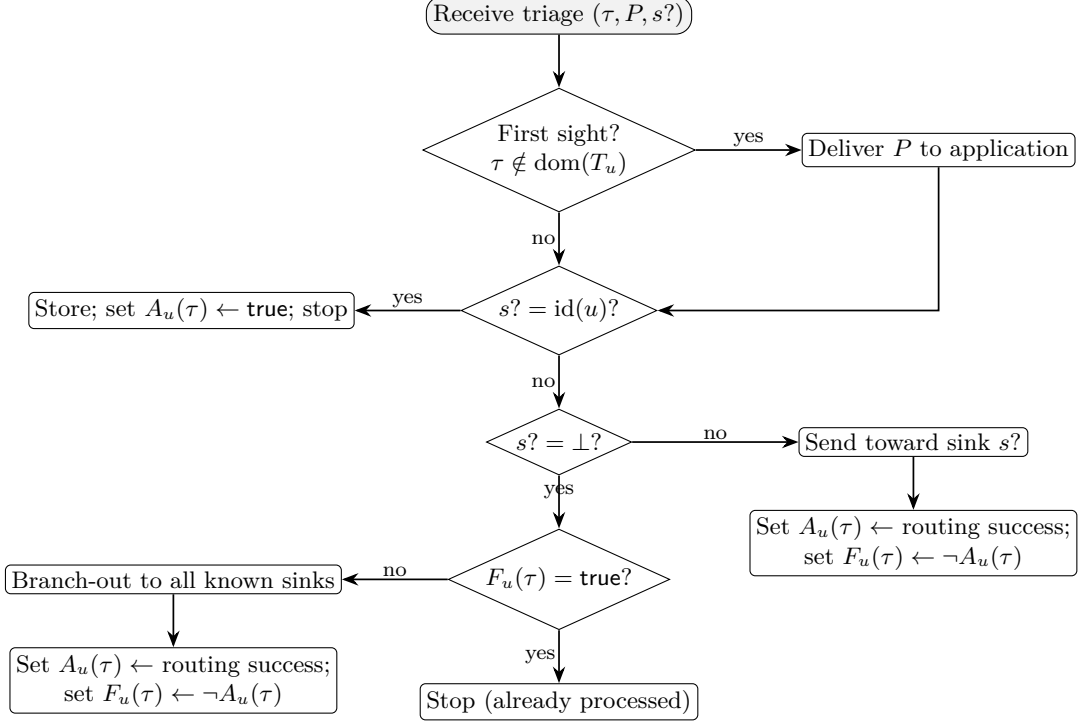


Figure 3: Triage receive/forward logic, reflecting the split between sink-addressed triages ($s? \in \Sigma^*$) and non-addressed triages ($s? = \perp$), and the role of flags $A_u(\tau)$ and $F_u(\tau)$.

8 Parameters

Parameter	Default
Routing retry bound for a sink-directed triage $N_{\max}$	3
Sink removal timeout T_{rm}	10 min
Maximum chunks per logical message	256